

Chapter 54: Pulmonary Disorders — Revision Notes

Williams Obstetrics 26e, pp. 2489–2538 of the PDF. High-yield numbers are in **bold**. The eight tables are transcribed from the PDF images (they are missing from the extracted chapter text). All 4 figures are included from the PDF.

Big picture: Asthma and community-acquired pneumonia cause almost **10%** of nonobstetrical antepartum admissions. Pregnant women (especially in the 3rd trimester) tolerate lung disease poorly because ↓ FRC and ↑ O₂ demand make them quick to become hypoxemic, and the fetus suffers first.

1. Respiratory Physiology Recap (see Ch 4)

Parameter	Change in late pregnancy
Vital capacity, inspiratory capacity	↑ ~ 20%
Expiratory reserve volume	1300 → ~ 1100 mL
Tidal volume	↑ ~ 40% (progesterone-driven)
Minute ventilation	↑ 30–40% → PaO ₂ 100 → 105 mm Hg
CO ₂ production	↑ 30%
PaCO ₂	40 → 32 mm Hg (hyperventilation + ↑ diffusion capacity)
Residual volume	↓ ~20% (1500 → 1200 mL)
Chest wall compliance	↓ by ⅓
Functional residual capacity	↓ 10–25% (= ERV + RV)
FVC, peak expiratory flow	Progressively ↑ from 14–16 wk
Basal O ₂ consumption	↑ 20 → 40 mL/min in 2nd half

- Net effect: **deeper, not more frequent, breathing.**

2. Asthma

Pathophysiology

- Chronic inflammatory airway syndrome with a strong hereditary component (**chromosome 5q** gene clusters: cytokines, β-adrenergic and glucocorticoid receptors, T-cell receptor).
- Morbidity higher in Black than white women.
- **Hallmarks: reversible airway obstruction** — smooth muscle contraction, vascular congestion, tenacious mucus, mucosal edema. Eosinophils, mast cells, T cells → histamine, leukotrienes, prostaglandins, cytokines, IgE.
- ⚠️ **F-series prostaglandins and ergonovine exacerbate asthma** → avoid these uterotonics.

Clinical course

- Obstruction ↓ **FEV₁/FVC** and **PEFR**; worsening oxygenation is mainly from **V/Q mismatch**.
- Early: hyperventilation keeps PaO₂ normal but ↓ PaCO₂ → **respiratory alkalosis**.
- Late: fatigue → CO₂ retention. **A "normal" PaCO₂ can be the first sign of retention** → respiratory failure.
- Even early stages are dangerous in pregnancy: ↓ FRC + ↑ pulmonary shunting → rapid hypoxemia.

Table 54-1. Classification of Asthma Severity

Component	Intermittent	Mild persistent	Moderate persistent	Severe persistent
Symptoms	≤2 d/wk	>2 d/wk, not daily	Daily	Throughout day
Nocturnal awakenings	≤2×/mo	3–4×/mo	>1×/wk, not nightly	Often 7×/wk
Short-acting β-agonist for symptoms	≤2 d/wk	≥2 d/wk, but not >1×/d	Daily	Several times daily
Interference with normal activity	None	Minor limitation	Some limitation	Extremely limited
Lung function	Normal between exacerbations			
FEV ₁	>80% predicted	≥80% predicted	60–80% predicted	<60% predicted
FEV ₁ /FVC	Normal	Normal	Reduced 5%	Reduced >5%

FEV₁ = forced expiratory volume in 1 second; FVC = forced vital capacity. From National Heart, Lung, and Blood Institute, 2007.

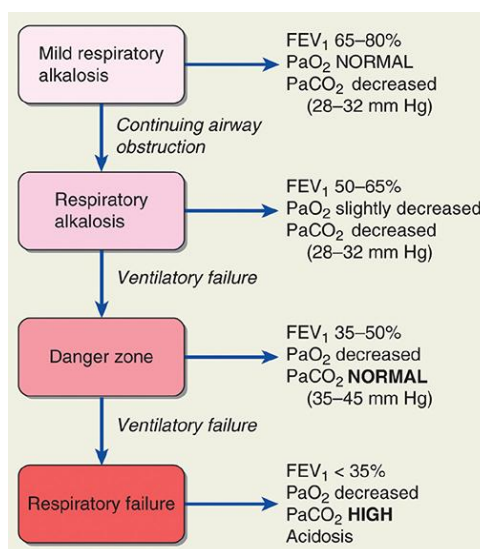


Figure 54-1. Clinical stages of asthma: a NORMAL PaCO₂ (35–45 mm Hg) with FEV₁ 35–50% is the "danger zone"; high PaCO₂ + acidosis = respiratory failure.

Stage	FEV ₁	PaO ₂	PaCO ₂
Mild respiratory alkalosis	65–80%	Normal	↓ (28–32 mm Hg)
Respiratory alkalosis	50–65%	Slightly ↓	↓ (28–32 mm Hg)
Danger zone	35–50%	↓	NORMAL (35–45)
Respiratory failure	<35%	↓	HIGH + acidosis

Effects of pregnancy on asthma

- Affects up to **8%** of gravidas.
- **Rule of thirds:** ~1/3 improve, 1/3 unchanged, 1/3 worsen (Gluck, >2000 women).
- Exacerbations track baseline severity (Schatz, 2003):

Baseline severity	Exacerbation	Admission
Mild	13%	2.3%
Moderate	26%	7%
Severe	52%	27%

- Intrapartum exacerbation: 20% (mild/moderate, Schatz) vs only 1% (Wendel). **18-fold ↑ risk after cesarean** vs vaginal delivery (Mabie).

Table 54-2. Maternal and Perinatal Outcomes in Women with Asthma

Outcome (asthmatics ^a)	Odds ratio (95% CI)
Maternal	
Hemorrhage	1.09 (1.03–1.16)
Gestational diabetes	1.10 (1.03–1.19)
Chorioamnionitis	1.12 (1.09–1.15)
Preeclampsia	1.14 (1.06–1.22)
Placental abruption	1.22 (1.09–1.36)
Placenta previa	1.30 (1.08–1.56)
ICU admission	1.34 (1.04–1.72)
Pulmonary embolism	1.71 (1.05–2.79)
Perinatal	
SGA neonates	1.10 (1.05–1.16)
Preterm delivery	1.17 (1.12–1.23)
Preterm PROM	1.18 (1.07–1.30)
Anomalies	1.48 (1.04–2.09)

^aCompared with outcomes in 206,468 nonasthmatic women. CI = confidence interval; ICU = intensive care unit; PROM = premature rupture of membranes; SGA = small for gestational age. Data from Baghlaf, 2019; Mendola, 2013, 2014.

- Findings are inconsistent: some large studies show ↑ pneumonia and cesarean but not perinatal mortality; spontaneous abortion may be slightly ↑.
- **Severe or poorly controlled asthma** drives risk: preterm birth **~2x** with severe asthma (MFMU). Baseline FEV₁ correlates directly with birthweight and inversely with gestational hypertension and preterm delivery.
- **Status asthmaticus** → respiratory arrest, pneumothorax, pneumomediastinum, acute cor pulmonale, arrhythmias. Mortality rises steeply once **mechanical ventilation** is needed.

Fetal effects

- **Maternal respiratory alkalosis causes fetal hypoxemia before maternal oxygenation is compromised:** ↓ uterine blood flow, ↓ venous return, **left shift** of the O₂ dissociation curve.
- Fetal response: ↓ umbilical flow, ↑ systemic and pulmonary vascular resistance, ↓ cardiac output. FGR ↑ with asthma severity.
- **The FHR is a monitor of maternal status:** return of moderate variability/accelerations = improving maternal oxygenation.
- Drugs: possible slight ↑ cleft lip/palate and autism (unconfirmed). **Up to half stop essential treatment at 5–13 wk** — a real concern.

Clinical evaluation

- Subjective severity correlates poorly with objective measures.

- Useful signs: labored breathing, tachycardia, **pulsus paradoxus**, prolonged expiration, accessory muscles. **Central cyanosis and altered consciousness = potentially fatal.**
- **Serial FEV₁ or PEF_R is the best measure of severity.** FEV₁ <1 L or <20% predicted = severe disease (hypoxia, poor response, high relapse).
- PEF_R correlates well with FEV₁ and is cheap; each woman should know her own asymptomatic baseline.

Management of chronic asthma

1. Patient education.
2. Avoid or control triggers (viral URIs, including colds). **Influenza and pneumococcal vaccination** encouraged.
3. Objective monitoring: PEF_R/FEV₁; fetal **serial growth ultrasound** ± antepartum testing.
4. Pharmacotherapy for baseline control and exacerbations; periodic medication review.
5. Treat related disorders: smoking, depression, rhinitis, GERD.
 - Moderate–severe asthma: record FEV₁ or PEF_R **twice daily**. Target **FEV₁ >80% predicted**; predicted **PEF_R 380–550 L/min** (use her own baseline).
 - **FeNO-guided** management → **50% fewer exacerbations.**

Agents

- **Mild intermittent: SABA as needed** is usually enough.
- **Persistent: inhaled corticosteroids (ICS)**, usually twice daily; step up from low- to medium-dose.
 - ICS ↓ hospitalizations by **80%** (Blais); at Parkland ↓ readmissions by **55%**.
 - As-needed **budesonide–formoterol** beat albuterol alone for mild asthma (nonpregnant) → now in guidelines.
- **Theophylline:** little benefit, many side effects; oral maintenance only if ICS + β-agonist inadequate.
- **Antileukotrienes** (zileuton, zafirlukast, montelukast): prevention only, **ineffective acutely**; ~half respond; less effective than ICS; limited pregnancy data.
- **Cromones:** mast cell stabilizers, ineffective acutely; mainly childhood asthma.
- **Vitamin D:** low levels linked to asthma severity; antenatal supplementation did **not** lower childhood asthma at 6 years (RCT).

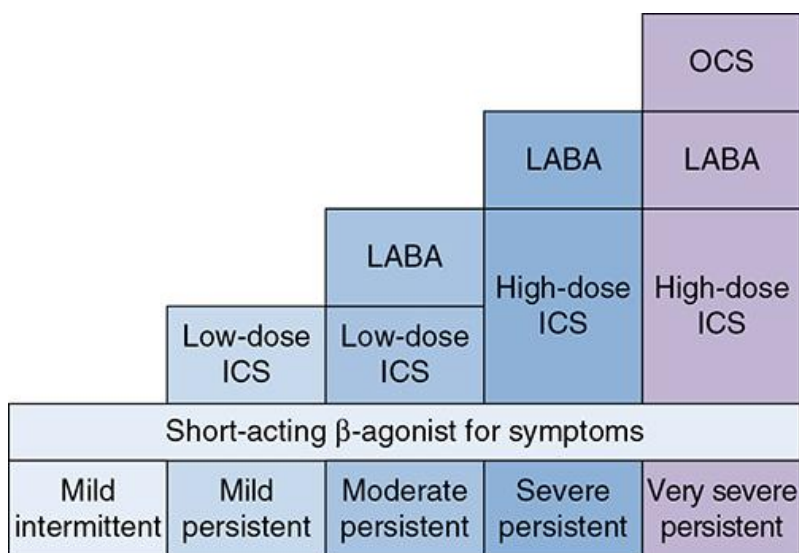


Figure 54-2. Stepwise therapy: SABA for all → low-dose ICS → + LABA → high-dose ICS + LABA → + oral corticosteroids.

Table 54-3. Chronic Asthma Treatment

SABA inhaler

Agent ^a	Brand	Dose per puff	Dosage
Albuterol	Proventil HFA, Ventolin HFA, ProAir HFA	90 µg	1–2 puffs every 4–6 hr
Levalbuterol	Xopenex HFA	45 µg	1–2 puffs every 4–6 hr

ICS inhaler^b — daily corticosteroid dosing

Agent ^a	Brand	Dose per puff	Low	Medium	High
Beclomethasone HFA	QVAR	40, 80 µg	80–240 µg	>240–480 µg	>480 µg
Budesonide DPI	Pulmicort	90, 180, 200 µg	180–600 µg	>600–1200 µg	>1200 µg
Ciclesonide HFA	Alvesco	80, 160 µg	80–160 µg	>160–320 µg	>320 µg
Flunisolide HFA	Aerospan	80 µg	320 µg	>320–640 µg	>640 µg
Fluticasone furoate DPI	Arnuity Ellipta	100, 200 µg daily	100 µg	NA	200 µg
Fluticasone propionate DPI	Flovent Diskus	50, 100, 250 µg	100–300 µg	>300–500 µg	>500 µg
Fluticasone propionate HFA	Flovent HFA	44, 110, 220 µg	88–264 µg	>264–440 µg	>440 µg
Mometasone DPI	Asmanex	110, 220 µg	110–220 µg	330–440 µg	>440 µg
Mometasone HFA	Asmanex HFA	100, 200 µg	200 µg	400 µg	>400 µg

LABA/ICS inhaler^b — daily dosing

Agent ^a	Brand	Dose per puff	Low	Medium	High
Budesonide/formoterol	Symbicort	80/4.5, 160/4.5 µg	320/18 µg	>320/18–640/18 µg	>640/18 µg
Fluticasone/salmeterol DPI	Advair Diskus	100/50, 250/50, 500/50 µg	100/50–350/100 µg	350/100–500/100 µg	>500/50 µg
Fluticasone/salmeterol HFA	Advair HFA	45/21, 115/21, 230/21 µg	100/50–250/50 µg	300/50–500/50 µg	>500/50 µg
Fluticasone/salmeterol DPI	AirDuo Respiclick	55/14, 113/14, 232/14 µg	110/28 µg	226/28 µg	464/28 µg
Fluticasone/vilanterol	Breo Ellipta	100/25, 200/25 µg daily	100/25 µg	NA	200/25 µg
Mometasone/formoterol	Dulera	110/5, 200/5 µg	NA	400/20 µg	800/20 µg

^aAgents in each group are arranged alphabetically. ^bTwice-daily dosing unless otherwise stated. DPI = dry powder inhaler; HFA = hydrofluoroalkane (aerosol propellant); ICS = inhaled corticosteroid; LABA = long-acting β -agonist; NA = not applicable; SABA = short-acting β -agonist. From American Academy of Allergy Asthma & Immunology, 2020; NHLBI, 2007; Therapeutic Research Center, 2017.

Management of acute asthma

- Same as nonpregnant, but the **threshold for hospitalization is much lower**.
- IV hydration; **O₂ by mask** to keep **PaO₂ >60 mm Hg** (preferably normal) and **SpO₂ 90–95%**.
- Baseline FEV₁/PEFR, continuous pulse oximetry, EFM as appropriate.
- ABG: interpret against pregnancy norms — **PaCO₂ >35 mm Hg with pH <7.35 = CO₂ retention** in a pregnant woman. Routine ABG didn't help most admissions (Wendel).
- **No antibiotics** unless pneumonia coexists.

Drugs

- **First line: short-acting β -agonist** (terbutaline, albuterol, etc.; SC, oral, inhaled or IV if severe).
- **Systemic corticosteroids early for all severe attacks** (onset takes hours): prednisone/prednisolone/IV methylprednisolone **30–45 mg daily for 5–10 days, no taper**. Parkland uses a higher IV dose.
- Add **nebulized ipratropium 0.5 mg every 20 min × 3**, then every 6–8 h.
- Severe: **IV magnesium sulfate** or theophylline.
- **Disposition:** FEV₁/PEFR **>70%** of baseline after β -agonist → consider discharge. **<70% after 3 doses** or obvious distress → **admit** (L&D, intermediate care or ICU).

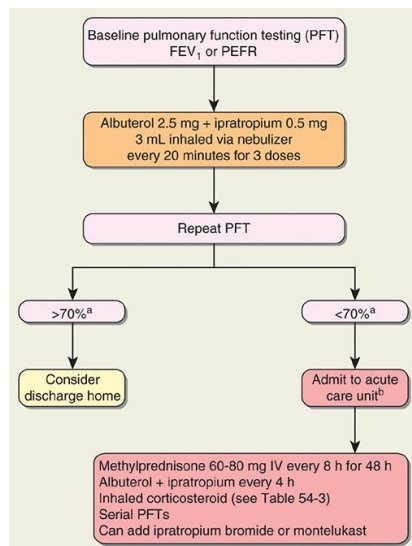


Figure 54-3. Parkland acute asthma protocol: albuterol 2.5 mg + ipratropium 0.5 mg neb q20 min × 3 → repeat PFT → >70% discharge, <70% admit for IV methylprednisolone 60–80 mg q8h × 48 h.

The printed flowchart reads "methylprednisone", an obvious typo for methylprednisolone.

Status asthmaticus and respiratory failure

- **Definition:** severe asthma **not responding after 30–60 min** of intensive therapy.
- In pregnancy, **intubate early** when status worsens despite aggressive treatment.
- Indications for ventilation: **fatigue, CO₂ retention, hypoxemia**.
- Refractory: **venovenous ECMO** has been used successfully.

Labor and delivery

- PEFr/FEV₁ on admission, then serially. **Continue maintenance drugs; stress-dose steroids not needed**.
- Ripening/induction: **oxytocin, PGE₁ or PGE₂** are fine.
- Analgesia: **fentanyl** (non-histamine-releasing) preferred over meperidine; **epidural ideal**.
- Cesarean: **conduction analgesia preferred** (intubation can trigger severe bronchospasm).
- **PPH:** oxytocin, misoprostol (PGE₁), dinoprostone (PGE₂). **⚠ PGF₂α (carboprost/Hemabate) is contraindicated** → bronchospasm.

Memory hook: "F is for Forbidden" — PGF₂α and ergonovine are avoided in asthma; PGE₁/PGE₂ are safe.

3. Acute Bronchitis

- Large-airway infection: **cough without pneumonitis**; mostly **viral** (influenza A/B, parainfluenza, RSV, coronavirus, adenovirus, rhinovirus).
- Cough lasts **10–20 days**, occasionally ≥1 month.
- Antibiotics: little evidence of benefit.
- Supportive: **dextromethorphan** and **guaifenesin** appear safe in pregnancy.

4. Pneumonia

- **Community-acquired pneumonia (CAP)** is the usual type in young women; outpatient-facility infections resemble hospital-acquired pneumonia.

- Pathogen found in only **38%** (CDC): viruses 23%, bacteria 11%, both 3%, fungi/protozoa 1%. **Half of bacterial isolates = *S. pneumoniae*.**
- Pneumonia = **4.2%** of antepartum nonobstetric admissions; also a common reason for postpartum readmission.
- Hypoxemia and acidosis are poorly tolerated by the fetus → may trigger **preterm labor**.
- **Any gravida suspected of pneumonia should have a chest radiograph.**

Bacterial pneumonia

- Risk factors: smoking and chronic bronchitis (*S. pneumoniae*, *H. influenzae*, *Legionella*), asthma, binge drinking, HIV.
- **Pregnancy itself does not predispose: 1.5 per 1000** deliveries (same as nonpregnant 1.47/1000).
- ≥Half viral, ~¼ bacterial (half of those pneumococcal). **CA-MRSA** → necrotizing pneumonia. Occasionally *Legionella*.

Diagnosis

- Cough, dyspnea, sputum, pleuritic pain; mild leukocytosis.
- **Chest radiograph is essential**; it does not predict the organism.
- Sputum culture, serology, cold agglutinins, bacterial antigens: **not routine**. Exception: **rapid nasal PCR for influenza A/B and COVID-19**. Procalcitonin didn't help.



Figure 54-4. CAP in pregnancy: rounded right middle lobe and left apical infiltrates.

Management

- PSI and CURB-65 guide admission but are **not validated in pregnancy**. **Parkland admits all gravidas with radiographic pneumonia.**
- **~20%** of admitted pregnant women need ICU/intermediate care. Severe pneumonia is a common cause of **ARDS** in pregnancy (12% of ventilated gravidas), though with lower mortality than other ARDS causes.

Table 54-4. Risk Factors for Early Deterioration in Community-Acquired Pneumonia

Risk factor
Respiratory rate $\geq 30/\text{min}$
Arterial O ₂ saturation $< 90\%$
Hypothermia: core temperature $< 36^\circ\text{C}$
Hypotension requiring aggressive fluid resuscitation
Multilobular infiltrates
Leukopenia: $< 4000/\mu\text{L}$
Thrombocytopenia: $< 100,000/\mu\text{L}$
Confusion/disorientation

Treatment

- Empirical. Healthy gravidas: pneumococcus, *Mycoplasma*, *Chlamydia* → **macrolide monotherapy** (erythromycin worked in 98/99 women).
- Severe disease (Table 54-4): **respiratory fluoroquinolone** OR **macrolide + β -lactam**. Macrolide resistance favors fluoroquinolones; fluoroquinolone teratogenic risk is low → **give if indicated**.
- **MRSA suspected**: add **vancomycin 15 mg/kg IV q12h** or **linezolid 600 mg q12h**.
- **During flu season add oseltamivir** routinely.

Table 54-5. Empirical Inpatient Antimicrobial Treatment for Community-Acquired Pneumonia in Pregnancy^a

Category	Regimen
Uncomplicated, otherwise healthy^{b, c}	Macrolides^c : clarithromycin 500 mg twice daily or azithromycin 500 mg daily (IV and oral forms)
<i>Plus</i>	Oseltamivir 75 mg orally twice daily for 5 d for suspected influenza A
Severe pneumonia^d	Respiratory fluoroquinolones : moxifloxacin 400 mg daily or levofloxacin 750 mg daily (IV and oral forms)
<i>or</i>	β-lactams : ampicillin/sulbactam 3 g IV q6h; ceftriaxone 1–2 g daily; ceftaroline 600 mg IV q12h; or cefotaxime 1–2 g q8h — coupled with a macrolide above
<i>Plus</i>	Oseltamivir, as above, for suspected influenza A

^aAntibiotics are discontinued after 5 to 7 days in those afebrile for 48 to 72 hours. ^bUse as inpatient or outpatient regimen. ^cDoxycycline may be given instead if postpartum. ^dSee Table 54-4 for criteria. IV = intravenous.

Course

- Improvement in **48–72 h**; fever resolves in **2–4 days** → switch to oral.
- Total **5–7 days**; must be **afebrile 48–72 h** and clinically stable before stopping.
- X-ray may take **up to 6 weeks** to clear. Persistent fever → serial radiographs.
- **~20% develop a pleural effusion; treatment failure up to 15%** → broaden cover, CT for **empyema** needing drainage.

Pregnancy outcome

- **~7%** need intubation; maternal mortality **0.8%**.
- Up to 1/3: **PROM and preterm delivery**; LBW **2x**; ↑ FGR, preeclampsia, cesarean.

Prevention: pneumococcal vaccines

- **PPSV23** (60–70% protective) and **PCV13**. **Not** recommended for healthy pregnant women.
- **Diabetes, chronic heart/lung/liver disease**: one lifetime **PPSV23**.
- **Immunocompromise, malignancy, CKD, cochlear implant, asplenia (e.g., sickle cell)**: **PCV13**, then **PPSV23 ≥ 8 wk later**, and again 5 years later.

Influenza pneumonia

- Influenza A and B: RNA viruses; winter epidemics; aerosolized droplets.
- **10% of pregnant women get influenza each year.** Onset **1–4 days** after exposure: fever, cough, myalgia, chills.
- Pregnant women are more likely to be **hospitalized and admitted to ICU**. Pneumonia in 12% (Parkland 2003–04).
- **H1N1 2009–10 pandemic:** 10% of admitted pregnant/postpartum women in ICU, 11% of those died; influenza = **12% of pregnancy-related deaths**. Risk factors: advanced pregnancy, smoking, chronic hypertension. ECMO may be lifesaving.
- **Primary influenza pneumonitis:** sparse sputum, interstitial infiltrates.
- **Secondary bacterial pneumonia** (strep/staph) is more common, **2–3 days after initial improvement; CA-MRSA case fatality 25%**.

Management

- **ACOG: treat ALL pregnant women with suspected or proven influenza**, regardless of rapid test (false negatives).
- Neuraminidase inhibitors (block viral release) — shorten illness by **1–2 days**, probably ↓ pneumonitis; low risk in pregnancy:

Drug	Dose
Oseltamivir (Tamiflu) — preferred	75 mg PO twice daily × 5 days
Zanamivir (Relenza)	10 mg inhaled twice daily × 5 days
Peramivir (Rapivab)	600 mg IV once over 15–30 min
Baloxavir (Xofluza)	Not recommended in pregnancy

Prevention

- **Influenza vaccine recommended** (ACOG, CDC), once each season: any **inactivated (IIV) or recombinant (RIV)** vaccine. ⚠ **Live attenuated (LAIV) is contraindicated.**
- Only ~half of pregnant women were vaccinated (2018–19).
- Gives the infant temporary protection; no link to birth defects or childhood effects at 5 years; a miscarriage association (one case-control) should not change practice.

Coronavirus

- **SARS-CoV (2002):** case fatality ~10%, **up to 40% in pregnancy**; placental fibrin deposition and fetal thrombotic vasculopathy.
- **MERS-CoV** (Saudi Arabia 2013): contained.
- **SARS-CoV-2 (COVID-19):** course similar to nonpregnant, but ↑ **preterm labor**. Care in a **negative-pressure airborne infection isolation room**. Vertical transmission unclear; isolate the newborn from other neonates, but **rooming-in is allowed** with maternal mask and hand hygiene.

Varicella pneumonia

- Pneumonitis in **5%** of gravidas with varicella (see Ch 67).

Pneumocystis pneumonia

- *P. jirovecii* in immunocompromised, especially **AIDS** — the **most frequent HIV-related disorder in pregnancy**.
- Dry cough, tachypnea, dyspnea, **diffuse interstitial infiltrates**. Diagnosis: **PCR of bronchoalveolar lavage**.
- Mortality in pregnancy **50%** (22 cases).
- **Treatment: TMP-SMX for 14–21 days** (alternative sulfadiazine + pyrimethamine) + **glucocorticoids** (↑ survival).
- **Prophylaxis:** 1 **DS TMP-SMX daily** if **CD4 <200/μL**, **CD4 <14%**, or an AIDS-defining illness (especially oropharyngeal candidiasis).

Fungal pneumonia

- Usually in HIV/immunocompromise; mild and self-limited otherwise.
- **Histoplasmosis, blastomycosis**: not more frequent or severe in pregnancy.
- **Coccidioidomycosis**: pregnancy may ↑ **dissemination**, especially if diagnosed in the **3rd trimester**. Maternal mortality 40% overall (20% since 1973). **Erythema nodosum = better prognosis**; mediastinal lymphadenopathy suggests dissemination.
- **Cryptococcosis**: usually meningitis in pregnancy.

Antifungal	Pregnancy use
Itraconazole	Preferred for most disseminated fungal infections; ideally avoid in 1st trimester
Amphotericin B	Used extensively, no embryofetal effects
Fluconazole, ketoconazole	Embryotoxic in large doses early → avoid in 1st trimester
Echinocandins (caspofungin, micafungin, anidulafungin)	Teratogenic in animals; no human pregnancy data

5. Tuberculosis

- **1/3 of the world** is infected with *M. tuberculosis*. In the US: elderly, urban poor, minorities, HIV; **immigrants >half of active cases** (usually reactivation).
- Infection contained and dormant in **>90%**; reactivates with immunocompromise.
- Active disease: cough with little sputum, low-grade fever, **hemoptysis, weight loss**; infiltrates, cavitation, mediastinal nodes. **AFB smear positive in ~2%** of culture-positive cases.
- Extrapulmonary: lymph node, pleural, GU, skeletal, meningeal, GI, miliary. **Silent endometrial TB → tubal infertility.**

Tuberculosis and pregnancy

- **7.1 per 100,000** pregnancy-related hospitalizations. Positive TST in half of foreign-born NYC prenatal patients; 21% of HIV-positive gravidas (Miami).
- **72% of active disease is pulmonary.**
- Untreated TB: LBW, preterm and preeclampsia **~2×**; **perinatal mortality ~10×**. **Treated TB → usually good outcomes.**

Table 54-6. Adverse Maternal and Perinatal Outcomes in 3384 Pregnancies Complicated by Active Tuberculosis

Outcome	Odds ratio ^a (95% CI)
Maternal	
Cesarean delivery	2.1 (1.17–3.79)
ARDS	2.8 (1.35–6.10)
Anemia	3.8 (2.21–6.71)
Pneumonia	8.4 (5.8–12.3)
Miscarriage	9.0 (4.93–16.7)
Prenatal admission	9.6 (2.3–40.6)
Perinatal	
Preterm birth	1.6 (1.2–2.4)
Low birthweight	1.7 (1.2–2.4)
Anomalies	1.8 (1.24–2.62)
Fetal distress	2.3 (1.2–4.5)
Mortality	4.2 (4.9–11.8)

^aCompared with women without tuberculosis. ARDS = acute respiratory distress syndrome. Data from El-Messidi, 2016; Sobhy, 2017. The printed CI for perinatal mortality (4.9–11.8) does not contain the point estimate 4.2; one of the values is a misprint in the source, reproduced as printed.

Extrapulmonary TB

- **5.8%**; worse outcomes and often **delayed diagnosis** (9 of 13).
- Renal/intestinal/skeletal: $\frac{1}{3}$ LBW. **Tuberculous meningitis: $\frac{1}{3}$ of mothers died.**
- Spinal TB → paraplegia (psoas abscess in 5%). Peritoneal TB can mimic ovarian carcinomatosis or degenerating leiomyoma; TB meningitis can mimic hyperemesis.

Diagnosis

- **Latent TB:**
 - **IGRA** (blood test, interferon- γ release to *M. tuberculosis* antigens absent from **BCG**) — increasingly preferred.
 - **TST:** PPD intermediate strength **5 TU** intradermally. **Positive = ≥ 5 mm** → evaluate for active disease with **chest radiograph**.
- **Active TB: nucleic acid amplification + mycobacterial culture.** Notify the **state health department** (directly observed therapy, DOT).

Treatment

Latent infection

- Nonpregnant <35 y: rifamycin-based regimen $\pm$ isoniazid for **3–4 months** (preferred), or **isoniazid 300 mg daily for 6–9 months**.
- **Pregnancy:** INH is safe, but most **delay treatment until after delivery** (some until 3–6 months postpartum, because of $\uparrow$ INH hepatitis risk postpartum). Monitor transaminases.
- Postpartum completion is poor (**10–18%**).
- **Treat antepartum (exceptions):**

Group	Why
Recent skin-test converters	5% develop active disease in the 1st year
TST-positive + exposed to active TB	0.5%/year
HIV infection	~10% annual risk of active disease

Active infection

- Nonpregnant (WHO): **isoniazid + rifampin + ethambutol + pyrazinamide × 6 months** until susceptibilities are known. **Add pyridoxine** (INH neuropathy). New 4-month INH/rifapentine/moxifloxacin/PZA regimen is noninferior.
- **Pregnancy: 2-month intensive phase INH + rifampin + ethambutol**, then **7-month continuation with INH + rifampin** (9 months total).
- Pyrazinamide if resistance to rifampin/rifabutin.
- **⚠️ Aminoglycosides (streptomycin, kanamycin, amikacin, capreomycin) are fetal ototoxins → contraindicated.**
- **Breastfeeding is allowed**; breast-milk drug levels are too low to treat the neonate.
- **HIV**: starting ART and TB drugs together → **IRIS**; start ART **2–4 wk after** TB therapy. Rifampin/rifabutin interact with some PIs and NNRTIs.

Transmission prevention

- Airborne precautions, **negative-pressure room**; patient wears a surgical mask; staff wear fitted **N95**.
- Noninfectious when: (1) **2 weeks of DOT**, (2) **3 consecutive AFB-negative smears**, or (3) clinical improvement (afebrile, resolving cough).

Neonatal tuberculosis

- Congenital TB: via placental bacillema **or** aspiration of infected secretions at delivery (~half each).
- Mimics other congenital infections: hepatosplenomegaly, respiratory distress, fever, lymphadenopathy. Often diagnosed by **postpartum endometrial biopsy** of the mother.
- Unlikely if the mother is treated before delivery or sputum is culture negative.
- **Separate the newborn** from a mother suspected of active disease. **Infectious mother → neonatal INH prophylaxis for 3–6 months.**
- Untreated infant of a mother with active TB: **50% risk of disease in the first year.**

6. Sarcoidosis

- Chronic multisystem disease: T-helper cells + phagocytes in **noncaseating granulomas**; genetic predisposition.
- Prevalence **20–60/100,000**; equal sex; **>10× more common in Black people**; onset at **20–40 y**.
- Lung most common (then skin, eyes, lymph nodes). **>90% have an abnormal chest radiograph** at some point; **interstitial pneumonitis** is the hallmark; half have permanent changes.
- **Mediastinal lymphadenopathy 75–90%**; uveitis 1/3; skin 1/3 (**erythema nodosum**). Sarcoid causes ~10% of erythema nodosum in women.
- **Diagnosis: biopsy**, preferably a lymph node.
- Prognosis good: **50% resolve untreated**; 20% have mild chronic dysfunction; **~5% die**.
- **Treatment: glucocorticoids** for symptomatic disease (usually after a several-month observation period unless respiratory symptoms prominent): **prednisone 20–40 mg/d**, taper to **<10 mg by 6 months**. Poor response → immunosuppressives, cytokine modulators.

Sarcoidosis and pregnancy

- Incidence **9.6 per 100,000 births**. Meningitis, heart failure, neurosarcoidosis reported.
- Treat active disease as in nonpregnant women; serial PFTs if severe; **prednisone** for symptomatic uveitis, constitutional or pulmonary symptoms.

Table 54-7. Maternal and Perinatal Adverse Outcomes in 678 Women with Sarcoidosis

Outcome	Odds ratio ^a (95% CI)
Preeclampsia	1.62 (1.18–2.22)
Preterm delivery	1.73 (1.40–2.15)
Deep-vein thrombosis	4.92 (1.58–15.33)
Eclampsia	5.27 (1.69–16.40)
Pulmonary embolism	6.68 (3.99–11.21)

^aCompared with 7,094,000 women without sarcoidosis. CI = confidence interval.

- Preeclampsia and preterm delivery **~2x**; eclampsia, DVT, PE **5–7x**.

7. Cystic Fibrosis

Pathophysiology

- **Autosomal recessive**; >2000 mutations in the **CFTR** gene (**chromosome 7q**, 230 kb) — a **chloride channel**.
- Thick secretions obstruct exocrine ducts: sinuses, lung, pancreas, liver, reproductive tract. **Sweat test**: ↑ Na⁺, K⁺, Cl⁻.
- Lung: mucus plugging → infection (*Pseudomonas*, *S. aureus*, *H. influenzae*) → bronchiectasis → fibrosis → V/Q mismatch → pulmonary insufficiency (usual cause of death).
- **Median predicted survival 37 years**; **~80%** of females reach adulthood.
- **ΔF508 (Phe508del)** homozygosity is one of the most severe; **90%** of those with clinical disease carry ≥1 ΔF508 allele.

Preconceptional counseling

- Women are **subfertile** (tenacious cervical mucus); endometrium/tubes functionally normal, ovaries don't express CFTR. IUI and IVF can work.
- Men: **CBAVD** → **98% infertile**.
- **~4%** of affected women conceive each year.
- **Screening: ACOG** — **offer carrier screening to all women** pregnant or planning pregnancy. CF is on newborn screening. CFTR2 website classifies variants.

Prenatal care

- **Outcome is inversely related to severity of lung dysfunction.**
- **~10% have pregestational diabetes**; pregnancy insulin resistance → **GDM** after midpregnancy (a third needed insulin in one series); malnutrition from pancreatic insufficiency.
- **Pregnancy does not affect the course of CF** (matched for severity, no ↓ long-term survival).

Management

- **Prepregnancy counseling is imperative.** Watch for infection, diabetes, heart failure. **Serial PFTs**.
- **FEV₁ ≥50%** → usually tolerate pregnancy well.
- Postural drainage, bronchodilators, infection control; inhaled **recombinant DNase I**; inhaled **7% saline**.
- Nutrition, **calcium + vitamin D**, **pancreatic enzyme replacement**.
- CFTR correctors (tezacaftor–ivacaftor) help ΔF508 homozygotes — no pregnancy data.
- Infection (↑ cough and mucus): oral semisynthetic penicillin or cephalosporin for staph; **inhaled tobramycin or colistin** for *Pseudomonas*; serious infection → **admit immediately**.
- **Epidural** recommended for labor.

Pregnancy outcome

- **Prepregnancy FEV₁ <60% predicted** → ↑ preterm delivery, respiratory complications and **maternal death within a few years**.
- French registry: **~20% preterm, 30% FGR**; 17% of women died long-term.
- **Meconium peritonitis in 10–20% of affected fetuses**.
- Pregestational diabetes did not worsen outcomes (French registry). Perinatal outcomes surprisingly good.

Table 54-8. Odds Ratios for Maternal Complications in 1119 Pregnant Women with Cystic Fibrosis Compared with Controls

Complication	Odds ratio
Asthma	5
Thrombophilia	6
Diabetes	14
Acute kidney injury	16
Respiratory failure	30
Mechanical ventilation	32
Pneumonia	69
Death	125

Lung transplantation

- CF is a common reason for lung transplant. 10 pregnancies → 9 liveborns, but **maternal outcomes poor**: all 3 with rejection in pregnancy died of chronic rejection within 38 months.
- Median survival **10.5 years** after surviving the first year.

8. Carbon Monoxide Poisoning

- Normal COHb **1–3%** (nonsmokers); **smokers 5–15%** (+ polycythemia).
- Most common poisoning worldwide; poorly ventilated rooms with **space heaters**. Odorless, tasteless, high Hb affinity → displaces O₂.
- Sequelae: death, anoxic encephalopathy; **cognitive defects in up to half** after LOC or COHb **>25%**. Cortex, white matter, **basal ganglia** → later Parkinsonism.

Pregnancy and CO poisoning

- Endogenous CO production **almost doubles** in pregnancy. The mother is not more susceptible, but **the fetus tolerates it poorly**.

COHb (maternal)	Effect
5–20% (chronic)	Headache, weakness, dizziness, visual impairment, palpitations, N/V
30–50% (acute)	Impending cardiovascular collapse
>50%	May be fatal

- **HbF binds CO more avidly** → fetal COHb **10–15% higher** than maternal.
- **COHb half-life: 2 h mother vs 7 h fetus**. The fetus can be hypoxic before maternal levels rise much.
- Embryonic exposure → anomalies; later exposure → **anoxic encephalopathy**.
- Affected FHR: ↑ baseline, ↓ variability, absent accelerations/decelerations.

Treatment

- **Immediate 100% O₂** + supportive care for all.
- **Hyperbaric O₂** reduced cognitive defects (6 wk and 1 y) in adults; **recommended in pregnancy for "significant" exposure** — some use **maternal COHb >15–20%**. Example: 2.5 atm, 100% O₂ for 90 min at 21 wk → healthy term baby.

Quick-Recall: Drugs in Pregnancy with Lung Disease

Situation	Use	Avoid / caution
Chronic asthma	SABA prn; ICS (budesonide, etc.); LABA add-on; budesonide-formoterol prn	Theophylline (side effects); antileukotrienes/cromones not for acute attacks
Acute asthma	Albuterol + ipratropium nebs; early systemic steroids; IV MgSO ₄	Antibiotics unless pneumonia
Labor, asthmatic	Oxytocin, PGE ₁ /PGE ₂ , fentanyl, epidural	PGF₂α (Hemabate) , ergonovine , meperidine less preferred
CAP	Macrolide; severe → fluoroquinolone or β-lactam + macrolide; MRSA → vancomycin/linezolid	—
Influenza	Oseltamivir 75 mg bid × 5 d; IIV/RIV vaccine	Baloxavir; LAIV
PCP	TMP-SMX 14–21 d + steroids; prophylaxis if CD4 <200	—
Fungal	Amphotericin B; itraconazole after 1st tri	Fluconazole/ketoconazole early; echinocandins
TB (active)	INH + rifampin + ethambutol (2 mo) → INH + rifampin (7 mo) + pyridoxine	Aminoglycosides (streptomycin etc.) — fetal ototoxicity
Latent TB	INH after delivery (antepartum if converter, exposure, HIV)	—
CO poisoning	100% O ₂ ; hyperbaric O ₂ if significant (COHb >15–20%)	—

Key Numbers to Memorize

- Pregnancy: PaCO₂ **32 mm Hg**, PaO₂ **105 mm Hg**, tidal volume **+40%**, FRC ↓ **10–25%**
- Asthma in up to **8%**; **1/3 better / 1/3 same / 1/3 worse**; severe asthma exacerbation **52%**, admission **27%**
- Severe persistent asthma = FEV₁ **<60%**; severe disease FEV₁ **<1 L**; target FEV₁ **>80%**; PEFR **380–550 L/min**
- Acute asthma: keep **PaO₂ >60**, SpO₂ **90–95%**; **PaCO₂ >35 + pH <7.35 = retention**; discharge if **>70%**; status = no response in **30–60 min**
- Steroids **30–45 mg/d × 5–10 d**; ipratropium **0.5 mg q20 min × 3**
- Pneumonia **1.5/1000**; pathogen found in **38%**; ICU **20%**; ventilation **7%**; mortality **0.8%**; effusion **20%**; failure **15%**; treat **5–7 d**
- CAP deterioration: RR **≥30**, SpO₂ **<90%**, temp **<36°C**, WBC **<4000**, platelets **<100,000**
- Influenza **10%/year**; oseltamivir **75 mg bid × 5 d**; H1N1 = **12%** of pregnancy deaths
- PCP mortality **50%**; prophylaxis CD4 **<200** or **<14%**
- TB: pulmonary **72%**; perinatal mortality **~10×** untreated; TST **≥5 mm**; converters **5%** first year; HIV **10%/y**; therapy **9 months**; untreated neonate **50%**
- Sarcoid: **>90%** abnormal CXR; hilar nodes **75–90%**; **50%** remit; PE **OR 6.7**
- CF: survival **37 y**; FEV₁ **≥50%** tolerates pregnancy, **<60%** bad; meconium peritonitis **10–20%**; death **OR 125**
- CO: fetal COHb **10–15% higher**; half-life **2 h vs 7 h**; hyperbaric if **>15–20%**